



Marketing CRM Supporting Software Guide

LAKSHMI SAFETY EQUIPMENT PRIVATE LIMITED · how the CRM is built, how the parts connect, and what to check when something stops working · 07-10-2026

Contents

1. The CRM at a glance
2. Software it needs
3. How the parts connect
4. Folders on this computer
5. Settings file (.env)
6. Windows services, ports and firewall
7. First-time setup
8. Daily running and restart
9. Commands and tools
10. Mobile app
11. Backups
12. Updating and GitHub
13. SMS and mail
14. Troubleshooting
15. Security rules
16. Other documents

1. The CRM at a glance

What it is	A web CRM for sales targets, branch performance, requests (lead / enquiry / order / DC / sample), follow-ups, payment collection, reports, SMS, mail and HRM.	
Where it runs	On this office computer. Other computers and phones open it over the office Wi-Fi.	
This system link	<code>http://localhost/marketing_crm</code>	
Website link (office Wi-Fi)	<code>http://192.168.29.12/marketing_crm</code>	
Mobile link	<code>http://192.168.29.12/marketing_crm/app/</code>	
Health check page	<code>http://localhost/marketing_crm/health</code> shows in one page whether the database, folders and settings are OK.	
Project folder	<code>D:\AI PROJECT\SMS SYSTEM</code>	
Source code copy	GitHub: <code>https://github.com/mnivi2021-commits/LAKSHMI-MEDI-SHOP</code> (branch <code>main</code>)	

192.168.29.12 is this PC's address on the office Wi-Fi. If the router gives the PC a new address, the website and mobile links change. Ask the network person to **reserve** this address for the PC in the router.

2. Software it needs

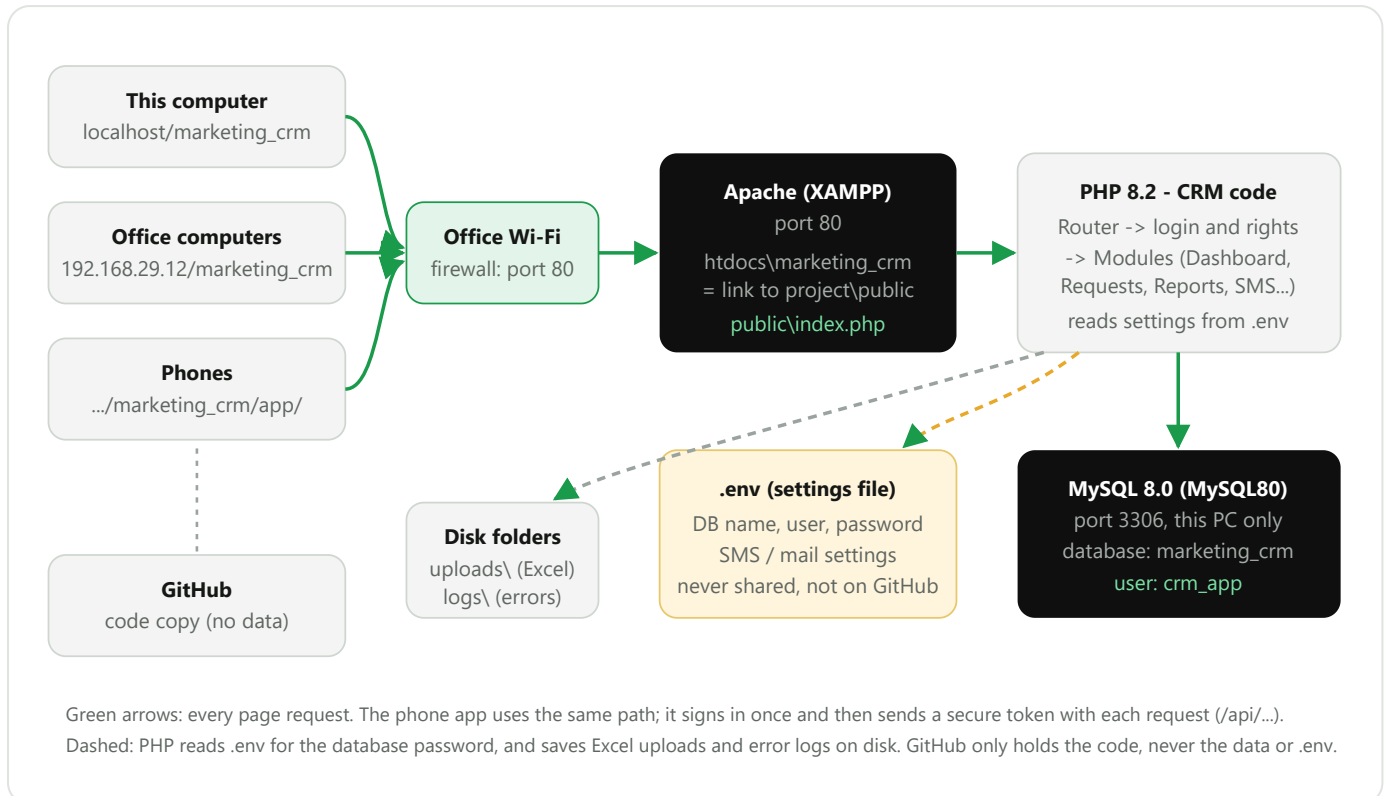
Software	Version on this PC	What it does for the CRM	Where it is
XAMPP - Apache	Apache 2.4.58	The web server. Receives every page request from browsers and phones and hands it to PHP.	C:\xampp\apache
XAMPP - PHP	PHP 8.2.12	Runs the CRM program code (all screens, rules, reports, Excel import, SMS, mail).	C:\xampp\php\php.exe
MySQL Server	MySQL 8.0.46 (Windows service MySQL80)	The database. Stores customers, sales, targets, collections, requests, users and settings.	C:\Program Files\MySQL\MySQL Server 8.0
Web browser	Chrome or Edge (current)	Used to open the CRM on computers and phones. Nothing is installed on user PCs.	-
Git	2.55	Keeps the code history and sends it to GitHub. Only needed for updates.	Git for Windows
Node.js + npm	Node 24.21, npm 11.19	Only for <i>building</i> the mobile app (Expo). Not needed to run the CRM day to day.	-
Expo / React Native	Expo 57, React Native 0.86	The mobile app source code (folder <code>mobile</code>). Its phone-browser build is already inside the CRM at <code>/app/</code> .	<code>mobile\</code>
PowerShell	Windows built-in	Runs the setup and Wi-Fi access scripts in <code>tools\</code> .	Windows

PHP extensions used

`pdo_mysql` (database), `mbstring` (Tamil / Unicode text, SMS length), `xml` / `xmlreader` / `simplexml` / `zlib` (reading Excel .xlsx files), `fileinfo` (checking uploaded files), `openssl` (secure tokens), `imap` (only if mail is fetched from a mailbox). All are part of XAMPP; `imap` is switched on in `C:\xampp\php\php.ini` only when needed.

No other paid software or internet service is needed for the CRM to run inside the office. SMS needs an SMS provider account only when you want real SMS (it is in test mode now).

3. How the parts connect



What happens when someone opens a page

1. The browser or phone sends the request over the office Wi-Fi to this PC on **port 80**. Windows Firewall allows it because of the rule "Marketing CRM (Apache, LAN only)".
2. **Apache** finds `C:\xampp\htdocs\marketing_crm`. That is a folder link pointing to `D:\AI PROJECT\SMS SYSTEM\public`, so the CRM files stay in the project folder.
3. Every request goes to `public\index.php`. **PHP** loads the settings from `.env`, checks the login session, checks the user's rights (role and branch / own data), then runs the screen's code.
4. The code reads and writes **MySQL** (database `marketing_crm`) as user `crm_app`, using the password in `.env`.
5. The finished page goes back to the browser. Excel uploads are saved in `uploads\`; errors are written to `logs\`.

If Apache is off, no link opens. If MySQL is off or the password in `.env` is wrong, pages open but show an error, and the health page shows **Database: FAIL**.

4. Folders on this computer

Folder (inside D:\AI PROJECT\SMS SYSTEM)	What is in it
public\	The only folder the web can see: <code>index.php</code> , styles, scripts, logo, and <code>app\</code> (phone version).
app\	CRM program code: <code>Core</code> (login, rights, database, security) and <code>Modules</code> (Dashboard, Requests, Followup, Reports, Sms, Mail, Hrm, Customers, Leads, Products, Imports, Access, Settings...). <code>Views</code> holds the screen layouts.
routes\	The list of every web address and which screen and permission it needs.
config\	Menu and app settings (menu order, which permission shows each menu).
database\	<code>schema.sql</code> (all tables), <code>base.sql</code> (roles, permissions, settings, SMS templates), <code>seed.sql</code> (demo data only), <code>migrations\</code> (changes for upgrades), <code>setup_user.sql</code> .
cli\	Commands run with PHP: install, migrate, create admin, health, data check, SMS sending, mail fetching.
tools\	<code>setup-crm.ps1</code> (one-step setup) and <code>allow-wifi-access.ps1</code> (firewall for office Wi-Fi).
mobile\	Source of the phone app (Expo). Build it only when the app changes.
uploads\	Excel files uploaded for import. Back this up.
logs\	Daily error and event logs <code>app-YYYY-MM-DD.log</code> . First place to look when something fails.
tests\	Automatic checks of the CRM. They erase the database they run on ; use only with a separate test database.
docs\	User guide, decisions, database guide, deployment guide and these PDFs.
screenshots\	Pictures of every screen.
.env	The settings file with passwords. Never share, never upload.

Outside the project: `C:\xampp\htdocs\marketing_crm` is only a link to `public\` . The database files are kept by MySQL in its own data folder; reach them only through MySQL (backups below), never by copying files.

5. Settings file (.env)

A plain text file in the project folder. PHP reads it on every request. Change it with Notepad, save, and the change applies at once (no restart).

Setting	Value now	Meaning
APP_URL	<code>http://localhost/marketing_crm</code>	Main address of the CRM.
APP_ENV / APP_DEBUG	<code>local / true</code>	For a real office server set <code>production</code> / <code>false</code> so error details are hidden.

Setting	Value now	Meaning
APP_TIMEZONE	Asia/Kolkata	Dates and "today" use Indian time.
DB_HOST / DB_PORT	127.0.0.1 / 3306	Where MySQL is (this PC).
DB_DATABASE	marketing_crm	Database name.
DB_USERNAME / DB_PASSWORD	crm_app / (secret)	CRM's own MySQL login. Made by <code>setup-crm.ps1</code> . Must match the MySQL user.
SESSION_LIFETIME	120	Minutes of no activity before sign-out.
LOGIN_MAX_ATTEMPTS / LOCKOUT	5 / 15 min	Wrong-password protection.
UPLOAD_MAX_MB	10	Largest Excel file allowed.
SMS_GATEWAY	mock	Test mode: SMS are recorded, not sent. See section 13.
MAIL_PROVIDER	manual	Mail is added by hand until a mailbox is connected.

Keep a safe copy of .env (pen drive or a password manager). Without it the CRM cannot reach its database after a reinstall. Never email it or upload it to GitHub (GitHub is set to ignore it).

6. Windows services, ports and firewall

Item	Name	Port	Start
Web server	Windows service Apache2.4 (installed by setup-crm.ps1)	80	Automatic with Windows
Database	Windows service MySQL80	3306 (this PC only)	Automatic with Windows
Firewall rule	"Marketing CRM (Apache, LAN only)"	80 inbound, office network only	Added once by allow-wifi-access.ps1

To see them: press **Win + R**, type `services.msc`, Enter. Both services should show **Running** and **Automatic**. Right-click > *Restart* fixes most "page not opening" problems.

MySQL port 3306 is **not** opened in the firewall: only this PC talks to the database. Never forward port 80 or 3306 from the router to the internet.

7. First-time setup (once per computer)

1. Install **XAMPP** (Apache + PHP 8.2) in `C:\xampp` and **MySQL Server 8.0** (remember the root password).
2. Copy or clone the project to `D:\AI PROJECT\SMS SYSTEM` and create `.env` from `.env.example`.
3. Make the web link (Command Prompt as administrator):

```
mklink /J C:\xampp\htdocs\marketing_crm "D:\AI PROJECT\SMS SYSTEM\public"
```

4. Open **PowerShell as administrator** and run:

```
cd "D:\AI PROJECT\SMS SYSTEM"
powershell -ExecutionPolicy Bypass -File tools\setup-crm.ps1
```

It creates the CRM database password in `.env`, the database and `crm_app` user (asks the MySQL root password), installs the tables, creates the Admin Head, sets Apache and MySQL to start with Windows, and opens office Wi-Fi access.

5. Open `http://localhost/marketing_crm/health`: everything should be **OK**. Sign in, change the password, then add users in **Access > Users**.

8. Daily running and after a restart

- Keep this PC switched on during office hours (a UPS is recommended). Apache and MySQL start by themselves.
- Users only need a browser and the link. Phones: open the mobile link and use *Add to Home screen*.
- After a power cut, wait one minute after Windows starts, then open the health page.

9. Commands and tools

Run in a Command Prompt or PowerShell inside the project folder. `php` means `C:\xampp\php\php.exe`.

Command	Use
<code>php cli\health.php</code>	Checks PHP, database, folders and settings; prints OK / FAIL.
<code>php cli\install.php</code>	Creates all tables on an empty database (adds <code>--seed</code> for demo data; <code>--fresh</code> erases everything first).
<code>php cli\migrate.php</code>	Applies database changes after an update. Safe to run any time.
<code>php cli\create-admin.php</code>	Creates the first Admin Head if none exists.
<code>php cli\verify-data.php</code>	Checks the data for mistakes after an Excel import.
<code>php cli\sms-dispatch.php</code>	Sends queued SMS campaigns (schedule every minute when real SMS is used).
<code>php cli\mail-sync.php</code>	Fetches mail from a connected mailbox (schedule every 5 minutes).
<code>tools\setup-crm.ps1</code>	One-step setup (section 7).
<code>tools\allow-wifi-access.ps1</code>	Lets office PCs and phones reach the CRM (firewall rule).

Scheduled jobs (only when real SMS / mail are switched on)

Windows **Task Scheduler** > Create Task > Action: `C:\xampp\php\php.exe`, Arguments: `"D:\AI PROJECT\SYSTEM\cli\sms-dispatch.php"`, Trigger: daily, repeat every 1 minute. Same for `mail-sync.php` every 5 minutes.

10. Mobile app

- **Phone browser version** is already built into the CRM at `/marketing_crm/app/`. It uses the same users and rights as the website.
- The app signs in through `/api/auth/login` and receives a secure token; every later request sends that token. Signing out, or disabling the user in Access, stops the token at once.
- Source code: `mobile\` (Expo). To rebuild the phone-browser version after app changes: `cd mobile`, `npm install`, `npm run build:web`.
- An installable Android app (APK / Play Store) needs an Expo account: `npx eas-cli build --platform android`, and a server reachable over HTTPS if used outside the office.

11. Backups

Make a daily copy of the database and keep one copy **off this PC** (pen drive, another PC, or cloud drive).

```
"C:\Program Files\MySQL\MySQL Server 8.0\bin\mysqldump.exe" --single-transaction -u crm_app -p marketing_crm > D:\CRM-Backup\crm_%date:~4%-%date:~3,2%-%date:~0,2%.sql
```

- Also copy the `uploads\` folder and the `.env` file.
- Put the command in a `.bat` file and run it daily with Task Scheduler.
- Every few months, test a restore into a separate test database.
- Backups contain customer data: do not email them or upload them to GitHub.

Restore

```
"C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -p marketing_crm < D:\CRM-Backup\crm_YYYY-MM-DD.sql
```

12. Updating and GitHub

GitHub keeps every version of the code (not the data, not `.env`). To bring this PC up to the latest version:

```
cd "D:\AI PROJECT\SMS SYSTEM"
git pull
C:\xampp\php\php.exe cli\migrate.php
C:\xampp\php\php.exe cli\health.php
```

Take a backup before updating. `migrate.php` only adds what is new and never deletes your data.

13. SMS and mail

- **SMS is in test mode** (`SMS_GATEWAY=mock`): messages are saved in SMS history but not sent. For real SMS in India: register the company, sender ID and every template on the provider's **DLT** portal, put the provider details in `.env`, connect the provider in `app\Services\Sms`, and send a test to your own number first.
- **Mail**: switch on `extension=imap` in `C:\xampp\php\php.ini`, restart Apache, create an *app password* for the mailbox, fill the `MAIL_IMAP_*` settings, then add the mailbox in **Mail > Mailboxes & rules**.

14. Troubleshooting

What you see	Likely reason	What to do
No link opens at all ("can't reach this page")	Apache is not running	<code>services.msc</code> > Apache2.4 > Start (or Restart). If the service is missing, run <code>tools\setup-crm.ps1</code> .
This PC works, other PCs / phones do not	Firewall rule missing, phone not on office Wi-Fi, or the PC's address changed	Run <code>tools\allow-wifi-access.ps1</code> as administrator. Check the address with <code>ipconfig</code> (IPv4). Reserve the address in the router.
"Access denied - check DB_USERNAME / DB_PASSWORD"	The CRM's MySQL login is missing or the password in <code>.env</code> does not match	Run <code>tools\setup-crm.ps1</code> again (it resets the <code>crm_app</code> password to the one in <code>.env</code>).
Health page: Database FAIL, "can't connect"	MySQL80 service stopped	<code>services.msc</code> > MySQL80 > Start.
"Apache won't start" / port 80 in use	Another program (Skype, IIS, another XAMPP) uses port 80	Close that program or stop IIS (<code>services.msc</code> > World Wide Web Publishing Service > Stop, Disabled).
A page shows "Something went wrong" with a reference number	A program error; details are logged	Open <code>logs\app-YYYY-MM-DD.log</code> and search for the reference number.
Forgot a user's password	-	Admin Head: Access > Users > Reset password .
Admin Head password lost and no other admin	-	Ask a technical person to reset it in MySQL, or restore from backup.
Menu missing for a user	Their role does not have that right	Admin Head: Access > Roles (or per-user rights).
SMS "sent" but not received	SMS test mode	See section 13.
Excel upload rejected	Wrong columns or file over 10 MB	Use the template from Excel Upload; the screen lists each wrong row.

15. Security rules

- Everyone has their own login. Never share the Admin Head login; never use the demo logins on real data.
- Rights are checked by the server on every request, not just by hiding menus.
- Keep `.env`, backups and the MySQL root password private. The CRM uses its own limited MySQL user, never root.
- Only use the CRM inside the office network unless it is moved to a proper server with HTTPS.
- Install Windows updates and keep an antivirus on this PC.

16. Other documents

docs\USER_GUIDE.md	How to use every screen.
docs\DECISIONS.md	Business rules and choices (targets, access, follow-up, SMS...).
docs\DATABASE.md	Tables and how dashboard figures are calculated.
docs\DEPLOYMENT.md	Moving to a hosted server, HTTPS, scheduled jobs, backups.
docs\LAKSHMI_CRM_Links.pdf	The three links on one page.
docs\LAKSHMI_CRM_Review.pdf	Pictures of every screen.